

Calculus Review

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1 Limits

$$\lim_{x \rightarrow a} f(x) = L$$

Intuitive understanding: As x approaches a infinitely, $f(x)$ approaches L infinitely.

2 Derivatives

Definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

3 Fundamental Theorem of Calculus

If f is continuous on $[a, b]$ and F is an antiderivative of f , then:

$$\int_a^b f(x) dx = F(b) - F(a)$$